

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2023

Course: B. Tech. Branch: Computer Engineering/CSE Semester: VII

Subject Code & Name: Blockchain Technology (BTCOE704C)

Max Marks: 60

Date: 09/01/2024

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Draw and explain layered architecture of Blockchain system.	(Understand)	6
B) What is Hash function? Explain its characteristics.	(Understand)	6
C) What Are the 4 Different Types of Blockchain Technology? Explain with advantages and disadvantages of each type.	(Apply)	6
Q.2 Solve Any Two of the following.		12
A) What is Bitcoin? What are the advantages and disadvantages of Bitcoin as Currency?	(Apply)	6
B) What is Double spending explain with suitable example.	(Remember)	6
C) Write a difference between PoW and PoS.	(Understand)	6
Q. 3 Solve Any Two of the following.		12
A) What the various steps of executing Smart Contract explain with suitable example?	(Understand)	6
B) Write a difference between permission and permissionless Blockchain.	(Understand)	6
C) Explain RAFT consensus algorithm.	(Apply)	6
Q.4 Solve Any Two of the following.		12
A) What is KYC? Explain the Shared KYC model based on Blockchain?	(Understand)	6
B) What is the blockchain mortgage process? Explain it with a suitable example. The advantages of blockchain in mortgages.	(Understand)	6
C) What is Supply Chain Financing? Give an example with justification.	(Apply)	6
Q. 5 Solve Any Two of the following.		12
A) Draw and explain the architecture of Hyperledger Fabric.	(Understand)	6
B) Explain the procedures needed in creating a smart contract based on Ethereum.	(Apply)	6
C) Explain the Ripple Protocol Consensus Algorithm (RPCA)	(Understand)	

***** End *****